Name : Kunal Motwani **Batch : T21** **Roll No : 72**

EXPERIMENT 9

Aim :

To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers

Theory :

Docker is a widely used containerization platform that allows developers to package applications along with their dependencies into isolated environments known as containers. These containers are lightweight and portable, enabling applications to run consistently across different systems. Unlike traditional virtual machines, Docker containers share the host system's operating system kernel, which makes them more efficient, faster to start, and less resource-intensive.

Docker operates using a client-server architecture. The Docker client acts as the interface through which users issue commands, while the Docker daemon (server) processes these commands and manages Docker objects such as images, containers, networks, and volumes. The communication between the client and daemon ensures smooth execution of container-related operations.

The Docker ecosystem is built around several core components. Docker images are read-only templates that contain the application code, runtime, libraries, and dependencies required to run an application. These images are used to create containers, which are the running instances of images. Docker also utilizes registries, such as Docker Hub, where images are stored and can be shared or downloaded. The Docker daemon handles tasks like building images, running containers, and managing their lifecycle.

The container life cycle in Docker consists of multiple stages. Initially, a container is created from an image. Once created, it can be started to execute the application. During execution, containers can be paused and resumed without terminating the process. When required, containers can be stopped, restarted, or removed completely from the system. This flexible life cycle allows efficient resource management and easy control over application execution.

By using various Docker commands, users can build images, pull images from registries, run containers, inspect their status, and manage them effectively. Docker simplifies the development and deployment process by ensuring consistency across different environments, making it an essential tool in modern DevOps practices.

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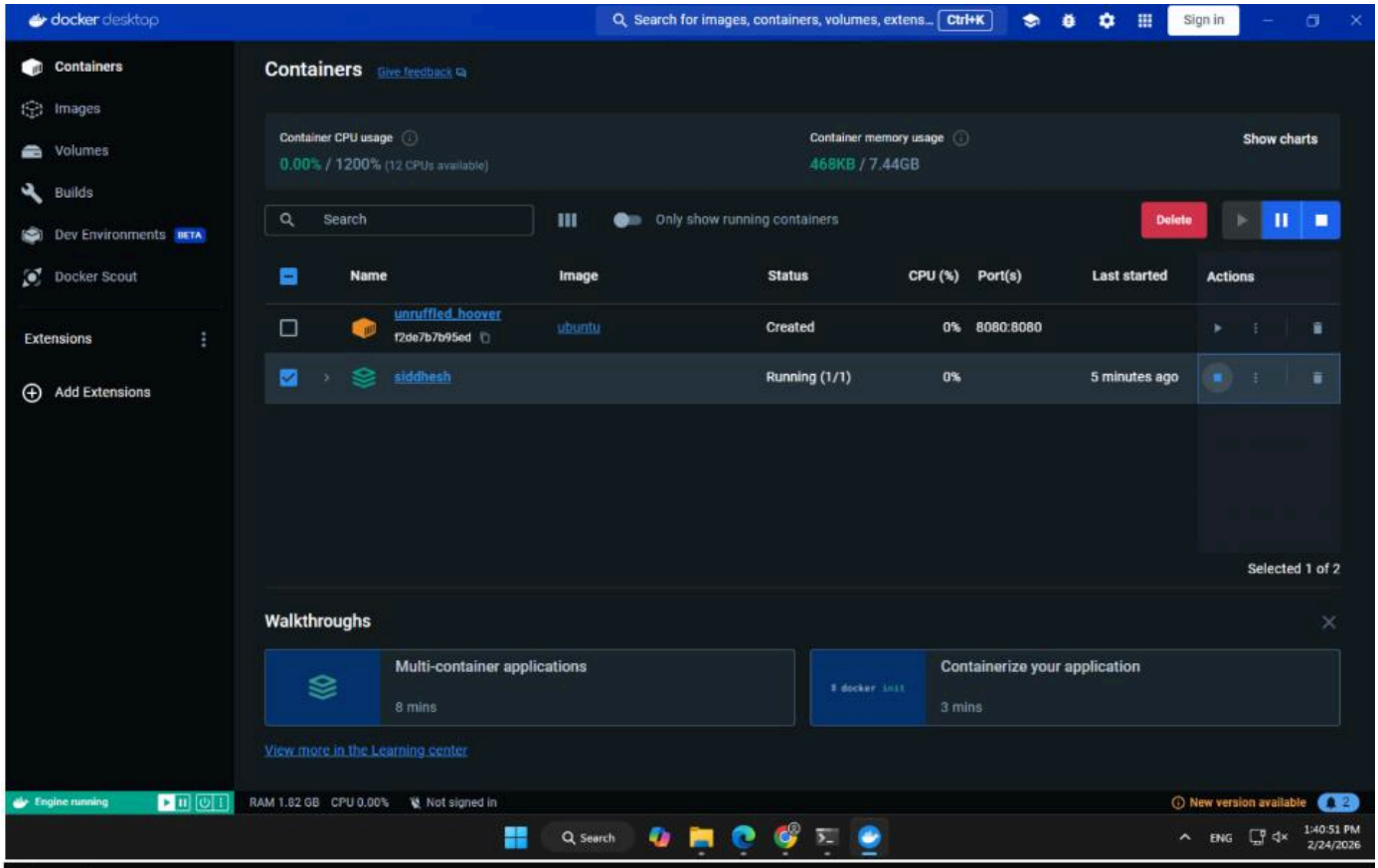


```
C:\Users\Lenovo>docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build       Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login       Log in to a registry
  logout      Log out from a registry
  search      Search Docker Hub for images
  version     Show the Docker version information
  info        Display system-wide information

Management Commands:
  builder     Manage builds
  buildx*     Docker Buildx (Docker Inc., v0.12.1-desktop.4)
  compose*    Docker Compose (Docker Inc., v2.24.6-desktop.1)
  container   Manage containers
  context     Manage contexts
  debug*      Get a shell into any image or container. (Docker Inc., 0.0.24)
  dev*        Docker Dev Environments (Docker Inc., v0.1.0)
  extension*  Manages Docker extensions (Docker Inc., v0.2.22)
  feedback*   Provide feedback, right in your terminal! (Docker Inc., v1.0.4)
  image       Manage images
  init*       Creates Docker-related starter files for your project (Docker Inc., v1.0.1)
  manifest    Manage Docker image manifests and manifest lists
  network     Manage networks
  plugin      Manage plugins
  sbom*       View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc., 0.6.0)
  scout*      Docker Scout (Docker Inc., v1.5.0)
  system      Manage Docker
  trust       Manage trust on Docker images
  volume      Manage volumes
```

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```
C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker search ubuntu
NAME                            DESCRIPTION                                     STARS     OFFICIAL
ubuntu                          Ubuntu is a Debian-based Linux operating sys…   17789     [OK]
ubuntu/squid                    Squid is a caching proxy for the Web. Long-t…   124
ubuntu/nginx                    Nginx, a high-performance reverse proxy & we…   139
ubuntu/cortex                   Cortex provides storage for Prometheus. Long…   4
ubuntu/bind9                    BIND 9 is a very flexible, full-featured DNS…   117
ubuntu/kafka                    Apache Kafka, a distributed event streaming …   60
ubuntu/zookeeper                ZooKeeper maintains configuration informatio…   14
ubuntu/apache2                  Apache, a secure & extensible open-source HT…   102
ubuntu/prometheus               Prometheus is a systems and service monitori…   78
ubuntu/mysql                    MySQL open source fast, stable, multi-thread…   72
ubuntu/dotnet-aspnet            Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/jre                      Chiseled Java runtime based on Ubuntu. Long-…   24
ubuntu/postgres                 PostgreSQL is an open source object-relation…   42
ubuntu/python                   A chiselled Ubuntu rock with the Python runt…   31
ubuntu/redis                    Redis, an open source key-value store. Long-…   24
ubuntu/dotnet-runtime           Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/dotnet-deps              Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                  Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                Memcached, in-memory keyvalue store for smal…   6
ubuntu/prometheus-alertmanager  Alertmanager handles client alerts from Prom…   10
ubuntu/cassandra                Cassandra, an open source NoSQL distributed …   3
ubuntu/mlflow                   MLFlow: for managing the machine learning li…   6
ubuntu/loki                     Grafana Loki, a log aggregation system like …   2
ubuntu/mimir                    Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/opentelemetry-collector  This is a rock for the OpenTelemetry Collect…   0
```

```
C:\Users\Lenovo>docker search --filter=stars=1000 ubuntu
NAME      DESCRIPTION                                     STARS     OFFICIAL
ubuntu    Ubuntu is a Debian-based Linux operating sys…   17789     [OK]
```

```
C:\Users\Lenovo>docker search mysql
NAME                  DESCRIPTION                                     STARS     OFFICIAL
mysql                 MySQL is a widely used, open-source relation…   16070     [OK]
bitnami/mysql         Bitnami Secure Image for mysql                  147
circleci/mysql        MySQL is a widely used, open-source relation…   32
bitnamicharts/mysql   Bitnami Helm chart for MySQL                    0
cimg/mysql                                                            3
ubuntu/mysql          MySQL open source fast, stable, multi-thread…   72
google/mysql          MySQL server for Google Compute Engine          26
linuxserver/mysql     A Mysql container, brought to you by LinuxSe…   48
elestio/mysql         Mysql, verified and packaged by Elestio         3
alpine/mysql          mysql client                                    7
docksal/mysql         MySQL service images for Docksal - https://d…   0
eclipse/mysql         Mysql 5.7, curl, rsync                          3
ilios/mysql           Mysql configured for running Ilios              1
datajoint/mysql       MySQL image pre-configured to work smoothly …   2
ddev/mysql            ARM64 base images for ddev-dbserver-mysql-8.…   2
mirantis/mysql                                                        0
mysql/mysql-server    Optimized MySQL Server Docker images. Create…   1033
corpusops/mysql       https://github.com/corpusops/docker-images/     0
cleanstart/mysql      Secure by Design, Built for Speed, Hardened …   0
vulhub/mysql                                                          1
cbioportal/mysql      This repository hosts MySQL database images …   1
vitess/mysql          Lightweight image to run MySQL with Vitess      1
mysql/mysql-router    MySQL Router provides transparent routing be…   28
mysql/mysql-cluster   Experimental MySQL Cluster Docker images. Cr…   100
nasqueron/mysql                                                       1
```

```
C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker images
REPOSITORY                      TAG                                       IMAGE ID       CREATED        SIZE
ubuntu                          latest                                    bbdabce66f1b   13 days ago    78.1MB
ubuntu                          <none>                                    802541663949   6 months ago   78.1MB
ubuntu                          <none>                                    ca2b0f26964c   24 months ago  77.9MB
redis                           <none>                                    170a1e90f843   2 years ago    138MB
docker/dev-environments-default stable-1                                  7eb85f44c229   3 years ago    610MB
crazymax/linguist               7.20.0                                    dc5f9ae3be6a   3 years ago    72.2MB
docker/desktop-git-helper       5a4fca126aadcd3f6cc3a011aa991de982ae7000  efe2d67c403b   4 years ago    44.2MB

C:\Users\Lenovo>docker run -it ubuntu
root@715e3a4a7ed8:/# exit
exit

C:\Users\Lenovo>docker run -it ubuntu
root@2ddd2c7718ec:/# admin
bash: admin: command not found
root@2ddd2c7718ec:/# ^C
root@2ddd2c7718ec:/# exit
exit

C:\Users\Lenovo>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb700ef54: Pull complete
30973b009bab: Pull complete
85b0d820aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a390514d6767e8c62f29375d0577190be448f63b24f5f11d6b03f7bf18
Status: Downloaded newer image for httpd:latest
```

```
C:\Users\Lenovo>docker run -d mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
357926362d31: Pull complete
f3702ac323ca: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
af5bbf473e08d0b9a7264168926d9712c94a6d94497687fed89179d31e770ccd

C:\Users\Lenovo>docker run -d ubuntu
17ae4f0c3240c5b5ba81df3d23be8a1eef290d48684f89b47335567685514b7a

C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE                                   COMMAND              CREATED              STATUS              PORTS     NAMES
9f10642c64c6   httpd                                   "httpd-foreground"   About a minute ago   Up About a minute   80/tcp    wonderful_babbage
a388ed3e2ae7   docker/dev-environments-default:stable-1   "sleep infinity"     23 months ago        Up 12 minutes               siddhesh-app-1
```

Conclusion :

Thus, the concepts of Docker Architecture and Container Life Cycle were successfully understood and implemented. Docker was installed, and various commands were executed to manage images and interact with containers. This experiment demonstrated how Docker enables efficient application deployment, improves portability, and ensures consistency across different environments.